



MINING DEALER BEST PRACTICE

Connecting Rod Lifting Tool (Various Engines)

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Connecting Rod Lifting Tool (Various Engines)	0
1.0 Introduction	1
2.0 Best Practice Description.....	1
3.0 Implementation Steps	2
4.0 Benefits.....	3
5.0 Resources Required	3
6.0 Supporting Attachments / References	3
7.0 Related Best Practices	4
8.0 Acknowledgements.....	4

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4.05-170504-03

1.0 Introduction

The assembly and attachment process for the connecting rod is a critical part of an engine rebuild. Appropriate care and handling of this major component is critical for operation; therefore, it is important to give it the utmost care when completing an engine assembly or disassembly.

During the process, the connecting rods are stored in baskets and must be manually extracted by the technician to carry out the final assembly process. The technician must lean over the basket and pull the rods, as seen in the figure below. This repetitive process for each engine will eventually lead to discomfort for the technician and may even lead to injury down the road.

Implementation of this change is geared towards improving the handling of the engine connecting rods by improving the storage and packing process.



Before



After

This lifting tool aims to prevent hip and/or lower back pain for technicians, as well as minimize the amount of time spent operating lifting machines. Overall, the implementation of this tool will lead to an improvement in safety standards and improve production time.

2.0 Best Practice Description

Information on the oil pump lifting tool designed is as follows:

- Geometry: Curved cylindrical bar
- Appearance: Sickle-shaped
- Dimensions:
 - Bar thickness: 1-1/4" diameter
 - Bar + Cone Length: 32"
 - Weight: 5 kg
- Materials: Steel, aluminum, and Teflon

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Connecting Rod Lifting Tool	DATE 8/7/2018	CHG NO 00	NUMBER 4.05-170504-03

- Structure (Parts or Stages):
 - Solid, cylindrical bar made of steel
 - Teflon roller covering the straight side of the bar
 - Conical aluminum tip
 - 5/16" diameter pin securing the cone
- Personnel involved: Metal-mechanical technicians

More information on the dimensions and specifications can be found in the attachments section.

3.0 Implementation Steps

- Before using the developed lifting tool, be sure to check its condition (1)
- Place the lifting tool on the hook of the crane/lifting device being used. This will allow it to be moved to the storage area for the connecting rods
- Align the connecting rods in such a way that the tool can pass through the upper bore of the connecting rod. For this application, 3,500 motors were being used on the crane, which allowed for four connecting rods to be lifted at once (2)
- Raise the connecting rods and move them to the desired area. The repetitive motion for the technician is no longer required (3)



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Connecting Rod Lifting Tool

DATE
8/7/2018

CHG
NO
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NUMBER
4.05-170504-03

4.0 Benefits

The benefits offered by this improvement are as follows:

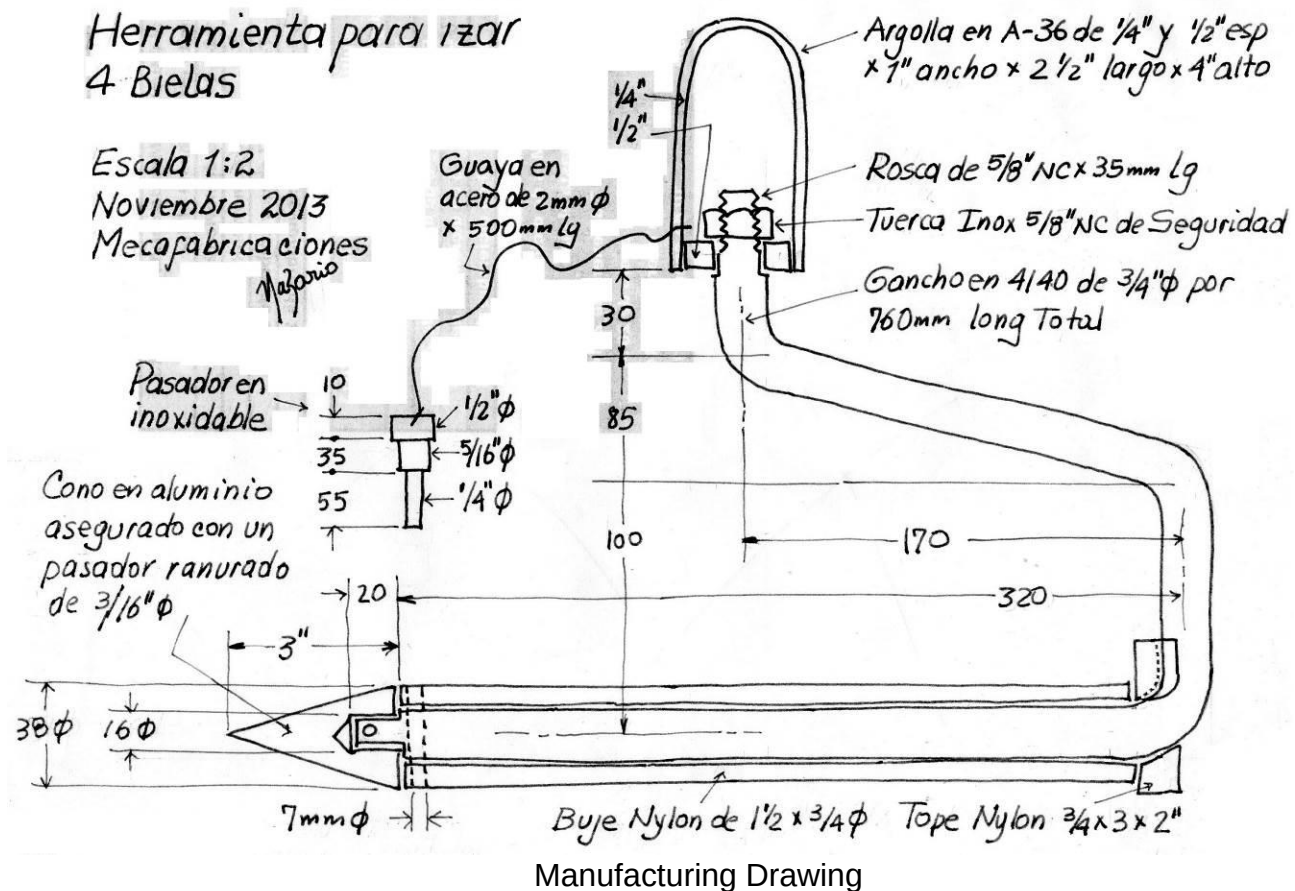
- Improved safety standards, primarily regarding lower back injuries and/or hernias
- Quicker fastening of the engine connecting rods without resorting to human force
- Improved production and energy consumed during the assembly process. From implementation, four engine connecting rods can be transported using the lifting tool instead of having to be carried one by one
- Improved maneuverability of the connecting rods is seen as they are placed on the work bench

5.0 Resources Required

To implement this best practice, the following resources are required:

- Materials as outlined in the *Best Practice Description* section. Total material investment totals USD \$240

6.0 Supporting Attachments / References



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Connecting Rod Lifting Tool

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8/7/2018

CHG
NO
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NUMBER
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7.0 Related Best Practices

None

8.0 Acknowledgements

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